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Habitat filtering masks the detection of phylogenetic and functional density dependence on seedling survival across seasons in a subtropical forest, Southwestern China

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Abstract

Background Negative density dependence and habitat filtering are crucial mechanisms driving species coexistence and community assembly. However, empirical evidence regarding how habitat filtering interacts with density dependence remains limited, particularly in terms of its influence on phylogenetic and functional density dependence in seedling dynamics. Seasonal variation in abiotic conditions (e.g., drought vs. high humidity) provides an opportunity to test whether habitat filtering masks or modifies density-dependent effects, as the strength of these mechanisms likely fluctuates with environmental stress. To address this gap, we assessed seasonal variation in seedling survival in response to phylogenetic distance and functional dissimilarity in a subtropical evergreen broad-leaved forest in Southwestern China, comparing models with and without habitat variables.

Results We found significant positive effects of phylogenetic distance and functional dissimilarity on seedling survival in the dry season, indicating that seedlings survived better when they were more phylogenetically or functionally distant from their neighbors. This provides evidence for both phylogenetic and functional negative density dependence (PNDD and FNDD). Habitat variables (elevation, convexity, and soil organic matter content) significantly predicted seedling survival only in the dry season. Furthermore, habitat filtering masked the effect of PNDD and FNDD at both community and species levels and influenced the detection of their interactive effects on seedling survival. Overall, we observed marked seasonal variation in the effects of both density dependence and habitat variables.

Conclusions This study demonstrates that habitat filtering obscures the detection of phylogenetic and functional density dependence in seedling survival and reveals a seasonally modulated link between neighborhood phylogenetic and functional structure and seedling survival. These findings highlight the importance of considering environmental context when assessing density-dependent mechanisms, advancing our understanding of forest community dynamics by showing that the strength and detectability of density dependence are contingent on both habitat conditions and seasonal abiotic stress.

Keywords Community assembly, Functional dissimilarity, Gaoligong Mountains, Neighborhood effects, Phylogenetic relatedness, Seasonal dynamics

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Introduction

Negative density dependence (NDD) is a vital stabilizing mechanism for species coexistence in forest ecosystems (Comita et al. 2014; Connell 1971; Janzen 1970; Queenborough et al. 2009; Wright 2002). Widespread across plant communities, this mechanism profoundly influences biodiversity maintenance and community assembly. Conspecific negative density dependence (CNDD)—a specific form of NDD—occurs when individuals experience reduced performance due to high densities of conspecific neighbors, attributed to host-specific natural enemies (e.g., pathogens, herbivores) and intraspecific competition for shared resources (Comita et al. 2014; Wright 2002). Numerous studies have documented CNDD across life stages and forest types, providing critical insights into patterns of survival, growth, and recruitment (e.g., Comita et al. 2009; Mensah et al. 2023; Metz et al. 2010; Yao et al. 2020).

Neighborhood dissimilarity can be characterized by taxonomic, phylogenetic, and functional differences among species. Previous studies have primarily focused on taxonomic distinctions, frequently classifying neighbors simply as conspecific or heterospecific (Luo et al. 2024; Queenborough et al. 2009; Yao et al. 2020). In species-rich subtropical forests, however, heterospecific neighbors dominate (Song et al. 2020; Wang et al. 2020), resulting in consistently low densities of conspecifics around focal seedlings. Under such conditions, the detectable effects of CNDD on plant performance are often weak (Chen et al. 2010; Wang et al. 2024a), suggesting that a simple conspecific and heterospecific dichotomy may overlook critical ecological dynamics. Furthermore, the balance between facilitative and competitive interactions is shaped by the diverse ecological differences among heterospecifics (Lebrija-Trejos et al. 2014).

As a result, phylogenetic and functional frameworks—which quantify neighborhood dissimilarity based on evolutionary relatedness and trait similarity, respectively—may help address this limitation. A growing body of empirical evidence indicates that NDD operates not only between conspecifics but also across phylogenetic and functional distance among neighbors (Lasky et al. 2014; Webb et al. 2006). Closely related neighbors decrease seedling performance (Metz et al. 2010; Pu and Jin 2018; Wang et al. 2024b; Webb et al. 2006; Zhu et al. 2015), a phenomenon termed phylogenetic negative density dependence (PNDD). This effect may be attributed to the role of phylogenetic dissimilarity in promoting resource partitioning and enhancing defense against natural enemies (Uriarte et al. 2010). Similarly, functional similarity among neighbors may also diminish plant performance (Chen et al. 2016; Lasky et al. 2014), leading to functional negative density dependence (FNDD). While PNDD is often attributed to resource partitioning and shared natural enemies among close relatives (Uriarte et al. 2010), it remains unclear whether FNDD operates through the same mechanisms. Alternatively, functional traits may capture additional dimensions of niche differentiation not reflected in phylogeny.

Nevertheless, the phylogenetic and functional trait approaches have inherent limitations that are complementary to one another. The phylogenetic approach may overlook density dependence mediated by phylogenetically labile functional traits, as the mechanisms driving NDD likely depend more directly on functional traits than on phylogenetic relatedness per se (Paine et al. 2012; Swenson 2013). Conversely, the functional trait approach is constrained by the feasibility of measuring all ecologically relevant traits (Cadotte et al. 2013). Although phylogenetically closely related and functionally similar species both tend to share limiting resources and specialized natural enemies (Vamosi et al. 2009), phylogenies and functional traits may capture different aspects of species' ecological strategies. Indeed, few studies have simultaneously examined how phylogenetic and functional density dependence jointly influence plant performance (Wang et al. 2020). Therefore, integrating both approaches is essential to determine whether closer phylogenetic relatedness and higher functional similarity lead to stronger negative neighborhood interactions among tree species.

Importantly, phylogenetic and functional distances are not independent: species that are closely related often share similar functional traits due to phylogenetic conservatism (Swenson 2013). However, phylogenetic distance may not always capture functional dissimilarity if traits are evolutionarily labile (Paine et al. 2012). Therefore, the interaction between phylogenetic and functional density dependence can provide deeper insights into the underlying mechanisms. For instance, a significant PNDD effect in the absence of FNDD might suggest that unmeasured, phylogenetically conserved traits (e.g., defense compounds) drive negative interactions. Conversely, if FNDD is strong only when phylogenetic distance is low, it may indicate that functional similarity amplifies competition or enemy sharing among close relatives. Disentangling these interactive effects is essential for understanding whether density dependence operates primarily through evolutionary history, functional traits, or their combination—a question central to predicting community responses to environmental change.

Seedling survival is critically regulated by surrounding environments, as conditions during the early establishment stage largely determine recruitment success (Comita et al. 2009; Johnson et al. 2017). In fact, a majority of new recruits persist for less than one

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season, leading to considerable intra-annual variation in seedling abundance (Lin et al. 2012). In seasonal forests, environmental conditions fluctuate markedly between dry and rainy seasons, potentially shifting the relative importance of abiotic and biotic drivers of seedling dynamics (Song et al. 2020; Wang et al. 2026). During the dry season, water stress is often the primary limiting factor, and habitat filtering—mediated by topography, soil properties, and light availability—may dominate by favoring species with drought-tolerant traits (Johnson et al. 2017; Poorter et al. 2006; Song et al. 2020). In contrast, the rainy season brings increased moisture and humidity, which can enhance the activity of fungal pathogens and insect herbivores, potentially intensifying negative density dependence (Lebrija-Trejos et al. 2023). Seasonal variation in temperature and moisture may further alter the strength of biotic interactions, such as competition and natural enemy pressure (Magee et al. 2021; Martini et al. 2022). Notably, even temporally invariant habitat factors like elevation and slope may exert seasonally varying effects on seedling survival by modulating microclimatic conditions (e.g., solar radiation, soil moisture retention) that differ between dry and rainy seasons (Song et al. 2020; Wang et al. 2026; Yao et al. 2020). Consequently, the drivers of seedling survival and the relative importance of habitat filtering versus density dependence are likely to shift seasonally. Quantifying these seasonal dynamics is essential for understanding how habitat conditions modulate the mechanisms underlying forest regeneration.

Central to these seasonal dynamics is the role of habitat filtering. Habitat filtering profoundly influences species distribution and performance, enabling focal seedlings and their neighbors to coexist and persist in the same suitable environment—a pattern consistent with positive density dependence (Kraft et al. 2015). However, the effects of PNDD and FNDD can reduce the survival of focal seedlings when they are surrounded by phylogenetically related or functionally similar neighbors. Thus, habitat filtering may modulate the detectable strength of negative density dependence (Chen et al. 2010; Johnson et al. 2017; Wu et al. 2016).

To address these knowledge gaps, we evaluate seasonal variations in the effects of habitat filtering and density dependence on seedling survival, specifically examining: (1) whether habitat variables mask the effects of phylogenetic and functional density dependence on seedling survival across dry and rainy seasons; (2) how this potential masking effect influences the interaction between phylogenetic and functional density dependence seasonally; and (3) whether such masking effects are consistent at both community and species levels.

Materials and methods

Study area

This study was conducted in a 4-ha (200 m × 200 m) forest dynamic plot within a subtropical mid-montane humid evergreen broad-leaved forest in Gaoligong Mountains (24° 50′ 9.8″–24° 50′ 17.3″ N, 98° 45′ 53.1″–98° 46′ 1.3″ E), Baoshan, Yunnan, Southwestern China (Fig. 1). The study area experiences a subtropical plateau monsoon climate, with an elevation ranging from 2135 to 2329.65 m and a mean annual temperature of 15.5 ℃. Mean annual precipitation is approximately 1650 mm, with over 87% occurring during the rainy season from May to October. During our study period (2020–2022), which encompassed two dry seasons (November–April) and two rainy seasons (May–October), the mean seasonal precipitation was 32.18 mm in the dry seasons and 226.86 mm in the rainy seasons (data from the Resource and Environmental Science Data Platform; https://www.resdc.cn). The soil is predominantly yellow–brown loam (Chai et al. 2014).

The entire plot was divided into 100 subplots (20 m × 20 m), each of which was further subdivided into 16 quadrats (5 m × 5 m). These quadrats served as the basic working units for tagging, mapping, measuring, and identifying all free-standing trees and shrubs with a diameter at breast height (DBH) ≥ 1 cm at 1.3 m above ground. A comprehensive forest survey was carried out in 2009–2010, recording a total of 10,546 individuals belonging to 95 species, 64 genera, and 35 families.

In November 2020, 100 seedling quadrats (2 m × 2 m) were established in the top-right corners of each subplot to monitor seedling dynamics. All woody seedlings (including trees, shrubs and lianas) with DBH < 1 cm and height ≥ 20 cm within each seedling quadrat were tagged and identified, and measured for height (from ground level to the tallest apical bud). Subsequent censuses were conducted in May and November for each year. During each census, newly recruited seedlings were measured for height, and their survival status was recorded. Across the four censuses from 2020 to 2022, a total of 936 seedlings were recorded, belonging to 56 species, 44 genera, and 23 families.

Conspecific neighbor densities

To quantify conspecific neighborhood effects on seedling survival, we calculated two variables: conspecific seedling density (S_con) and conspecific adult basal area (A_con). S_con was defined as the number of conspecific seedlings within the same seedling quadrat as the focal seedling. In contrast, A_con was calculated as the sum of basal areas of all conspecific trees within a 20 m radius of the focal seedling quadrat, weighted by the inverse of their distance (i.e., 1/distance) to the quadrat center.

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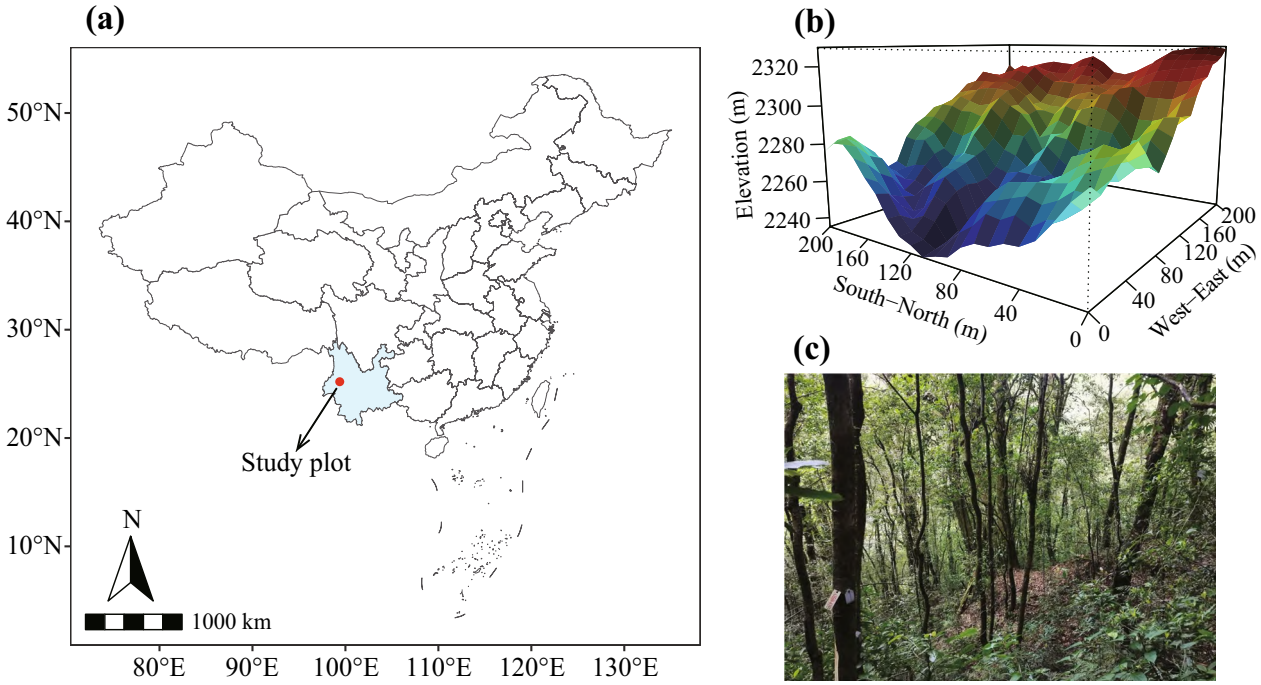


Fig. 1 Study area in Gaoligong Mountains, Yunnan, Southwestern China. **a** Geographic location; **b** topographic map of the 4-ha forest plot; **c** view of the mid-montane humid evergreen broad-leaved forest

This radius was chosen based on previous neighborhood modeling studies in the same plot (Wang et al. 2026). Our calculation of A_con assumed a linear relationship with basal area ($a=1$) and an inverse relationship with distance ($b=1$). However, previous studies have shown that incorporating exponents for basal area and distance can better capture the true competitive effects and avoid regression dilution bias (e.g., Detto et al. 2019; Song et al. 2020). While this parameterization is widely used and allows comparison with many previous studies, future work could explore whether optimizing these exponents alters the estimated strength of density dependence in this system.

Trait measurements

We focused on nine commonly measured leaf traits that reflect well-documented trade-offs in plant ecological strategies: specific leaf area (SLA), leaf area (LA), leaf dry-matter content (LDMC), leaf thickness (LT), chlorophyll content (Chl), and leaf carbon (LC), nitrogen (LN), phosphorus (LP), and potassium (LK) concentrations. SLA, Chl, and LDMC are associated with a plant's capacity for nutrient acquisition and utilization, whereas LA reflects light capture ability (Pérez-Harguindeguy et al. 2013). LN, LP, and LK are key determinants of photosynthetic efficiency and growth potential (Chave et al. 2009; Wright et al. 2004).

In May 2022, leaf samples were collected from all species recorded in the seedling censuses. For each species, we sampled five healthy adult individuals and collected five fully expanded leaves per individual. The sampled leaves were gently wiped clean, placed in plastic bags with silica gel for desiccation, and transported to the laboratory for subsequent trait measurements. We scanned fresh leaves using a digital scanner (Epson Perfection V39, Japan) and calculated LA (cm^2) from the images using ImageJ software (Schneider et al. 2012). After scanning, fresh leaf mass (g) was determined immediately using an electronic balance. Chl was assessed with a handheld chlorophyll meter (Konica Minolta SPAD-502, Japan) and averaged across three measurements per leaf. We used an electronic digital micrometer (Comecta, Barcelona, Spain) to measure LT at three random locations per leaf. Leaves were then oven-dried at 70 °C for at least 72 h before determining dry mass (g). LDMC was calculated as dry mass divided by fresh mass, and SLA was calculated as LA divided by dry mass. Prior to the determination of total C, N, P, and K concentrations, leaf samples were ground to a fine powder. We performed all leaf functional trait measurements following the protocols of Pérez-Harguindeguy et al. (2013).

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Phylogenetic and functional trait densities

To further quantify phylogenetic and functional density dependence among neighboring individuals, we first generated a phylogenetic tree for all species ($N = 109$) using the V.PhyloMaker 2 package (Jin and Qian 2022). Subsequently, we standardized trait data, built a functional trait dendrogram, and constructed a Euclidean distance matrix to represent functional distance among species. We then calculated the relative nearest taxon phylogenetic distance of seedling (S_NTPd') and adult (A_NTPd'), as well as the relative nearest taxon functional distance of seedling (S_NTFd') and adult (A_NTFd'), following the methods of Webb et al. (2006) and Wang et al. (2020). Detailed methodology is provided in Supplementary Method 1.

Habitat variables

In this study, we estimated habitat variables for each seedling quadrat, including soil properties, light availability, and topography.

Ten soil properties were measured from samples collected in May 2022 at a depth of 0–5 cm from three random locations within each seedling quadrat, for a total of 300 samples. The measured properties included pH, electrical conductivity (EC), available phosphorus (AP), available potassium (AK), soil organic matter (SOM), total nitrogen (TN), total phosphorus (TP), total potassium (TK), temperature, and moisture. Specifically, soil temperature and moisture were measured during each seedling census using a soil hygrothermograph (YDSC–A01). Soil pH, EC, AP, and AK were measured using a rapid nitrogen, phosphorus, and potassium meter (TRREC N01, Shandong Sean, China) at a depth of 0–5 cm. SOM was determined using the volumetric sulfuric acid–potassium dichromate oxidation method. TN was determined using the automatic nitrogen determination method, TP using the molybdenum antimony anticolorimetric method, and TK using the hydrofluoric acid and perchloric acid digestion flame photometer method.

Canopy openness served as a proxy for light availability and was assessed for each seedling quadrat in December 2021 using hemispherical photographs taken with a Nikon camera (Z5N1933) equipped with an F2.8 12-mm Senyo round fisheye lens. The camera was mounted on a tripod 1.3 m above the ground at the center of each quadrat. Photographs were taken early in the morning, late in the afternoon, or under uniformly cloudy skies. Canopy openness was calculated using Gap Light Analyzer v. 2.0 (Frazer et al. 1999) and ranges from 0% (completely closed) to 100% (completely open).

Topographic variables were derived from field surveys conducted in 2009–2010. Elevation of each subplot (20 m × 20 m) was calculated as the mean of the elevations at its four corners. Slope was defined as the mean angular deviation from the horizontal of each of the four triangular planes formed by connecting three of the four vertices of each quadrat. Convexity was calculated as the difference between the mean elevation of each subplot and the mean elevation of all its adjacent subplots.

To reduce dimensionality and collinearity among habitat variables in models, we performed principal components analysis (PCA) on soil properties and topography separately for the dry and rainy seasons. The first three principal components (PCA1–PCA3) were retained, collectively explaining 66.2% and 66.9% of the total variance in the dry and rainy seasons, respectively (Table S1). PCA1 was associated with higher elevation, convexity, and SOM; PCA2 represented lower SOM; and PCA3 was associated with lower convexity and slope but higher soil moisture. These principal components were subsequently used as predictors in the models.

Statistical analysis

To evaluate whether habitat filtering masks the effects of conspecific, phylogenetic, and functional density dependence on seedling survival, we used the lme4 package (Bates et al. 2015) with a binomial error distribution to construct two sets of generalized linear mixed models (GLMMs)—Model I and Model II—separately for the dry and rainy seasons. These models assessed the relationship between seedling survival (alive vs. dead) and a set of predictors comprising fixed and random effects. Fixed effects included seedling height, conspecific neighbor densities (S_con and A_con), heterospecific neighbor densities (S_NTPd', A_NTPd', S_NTFd', and A_NTFd'), and habitat factors. To explore the interactive effect of phylogenetic relatedness and functional similarity on seedling survival, we also incorporated the interaction terms (S_NTPd' × S_NTFd' and A_NTPd' × A_NTFd'). Random intercepts for seedling quadrat, year, and species identity were included to account for potential spatial autocorrelation, temporal variation, and interspecific differences, respectively. Seedling height was log-transformed (log H) to improve normality and reduce dispersion in model residuals. Prior to model fitting, all explanatory variables were standardized and centered to obtain standardized coefficients (Gelman and Hill 2006). To assess multicollinearity among predictors, we calculated variance inflation factors (VIFs) for all fixed effects using the car package (Fox and Weisberg 2019). All VIF values were below 5 (Table S2), indicating that multicollinearity was not a concern in our models (Zuur et al. 2010). These models are described in Supplementary Method 2.

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Two criteria were required for the NTPd'×NTFd' interaction to be considered strong. First, the interaction terms in the GLMMs had to be statistically significant ($P<0.05$). Second, following the framework of Laughlin et al. (2018), the slope of the relationship between NTPd' and logit-transformed survival had to switch signs across the range of the NTFd' gradient. This was assessed by computing the first partial derivative of Model II. We used the MuMIn package (Bartoń 2025) to extract the marginal (R^2_{mar}) and conditional (R^2_{con}) R-squared values for all models.

To confirm whether and how habitat filtering affects phylogenetic and functional density dependence among species, we added species-specific random slopes for these variables in Model I and Model II, respectively. The significance of these random slopes was evaluated using likelihood ratio tests (Bolker et al. 2009). If *P* values were less than 0.05, we inferred that the coefficients of neighborhood variables varied across species. We also used *t*-tests to compare the distributions of the species-specific coefficients of neighborhood variables between the varying-slope models with and without habitat variables.

We performed all analyses with R version 4.5.1 (R Core Team 2025).

Results

Habitat filtering masked the effect of phylogenetic and functional density dependence at the community level across seasons

Across both dry and rainy seasons, the coefficients of A_NTPd' were positive in Model I and Model II. This effect was significant in the dry season ($P<0.05$) but only marginally significant in the rainy season ($0.05<P<0.1$), indicating that seedlings survived better when surrounded by less closely related heterospecific neighbors—evidence of PNDD. The inclusion of habitat variables in Model II strengthened the estimated PNDD effects (Table 1; Fig. 2a, c).

Similarly, the influence of functional dissimilarity on seedling survival varied between models and across seasons. Specifically, in Model I, seedling survival was positively associated with S_NTFd' in the dry season, indicating that seedlings survived better when surrounded by functionally dissimilar neighbors—a pattern consistent with FNDD. However, this effect reversed to negative in the rainy season (Table 1; Fig. 2a, b). When habitat variables were included in Model II, the positive effect of S_NTFd' became stronger in the dry season, whereas no significant effect was detected in the rainy season (Table 1; Fig. 2c, d). In contrast, neither seedling height nor conspecific neighbor densities had a

Table 1 Coefficient estimates of explanatory variables for dry and rainy seasons in Model I (without habitat variables) and Model II (with habitat variables)

Explanatory variables	Dry season				Rainy season			
	Model I		Model II		Model I		Model II	
	Estimate	*P*	Estimate	*P*	Estimate	*P*	Estimate	*P*
log *(H)*	0.282	0.076	0.219	0.184	0.231	0.165	0.196	0.248
S_con	− 0.014	0.928	− 0.058	0.723	0.042	0.822	− 0.008	0.969
A_con	0.131	0.507	0.157	0.463	0.015	0.930	− 0.005	0.976
S_NTPd'	− 0.417	0.088	− 0.395	0.123	0.087	0.634	0.078	0.673
A_NTPd'	0.430	0.038	0.465	0.029	0.344	0.077	0.385	0.054
S_NTFd'	1.220	0.039	1.526	0.019	− 0.493	0.028	− 0.383	0.118
A_NTFd'	1.091	0.226	0.891	0.324	0.667	0.305	0.688	0.294
light			− 0.118	0.519			0.119	0.514
PCA1			− 0.405	0.031			− 0.259	0.188
PCA2			− 0.233	0.175			0.190	0.309
PCA3			0.173	0.308			− 0.011	0.946
S_NTPd'×S_NTFd'	− 1.180	0.024	− 1.246	0.032	0.586	0.055	0.622	0.041
A_NTPd'×A_NTFd'	1.099	0.137	1.113	0.127	0.613	0.248	0.723	0.177
R^2_{mar}	0.027		0.040		0.021		0.028	
R^2_{con}	0.027		0.040		0.038		0.043	

log *H* is the log-transformed height of the focal seedling. S_con and A_con are conspecific neighbor densities for seedlings and adults, respectively. Heterospecific neighborhood variables include the relative nearest taxon phylogenetic distance (S_NTPd' and A_NTPd') and functional distance (S_NTFd' and A_NTFd') to seedling and adult neighbors. Habitat variables comprise canopy openness (light) and the first three principal components (PCA1, PCA2, and PCA3) of soil properties and topography. Marginal R^2 (R^2_{mar}) and conditional R^2 (R^2_{con}) indicate the proportion of variance explained by fixed effects alone and by the entire model (fixed and random effects), respectively

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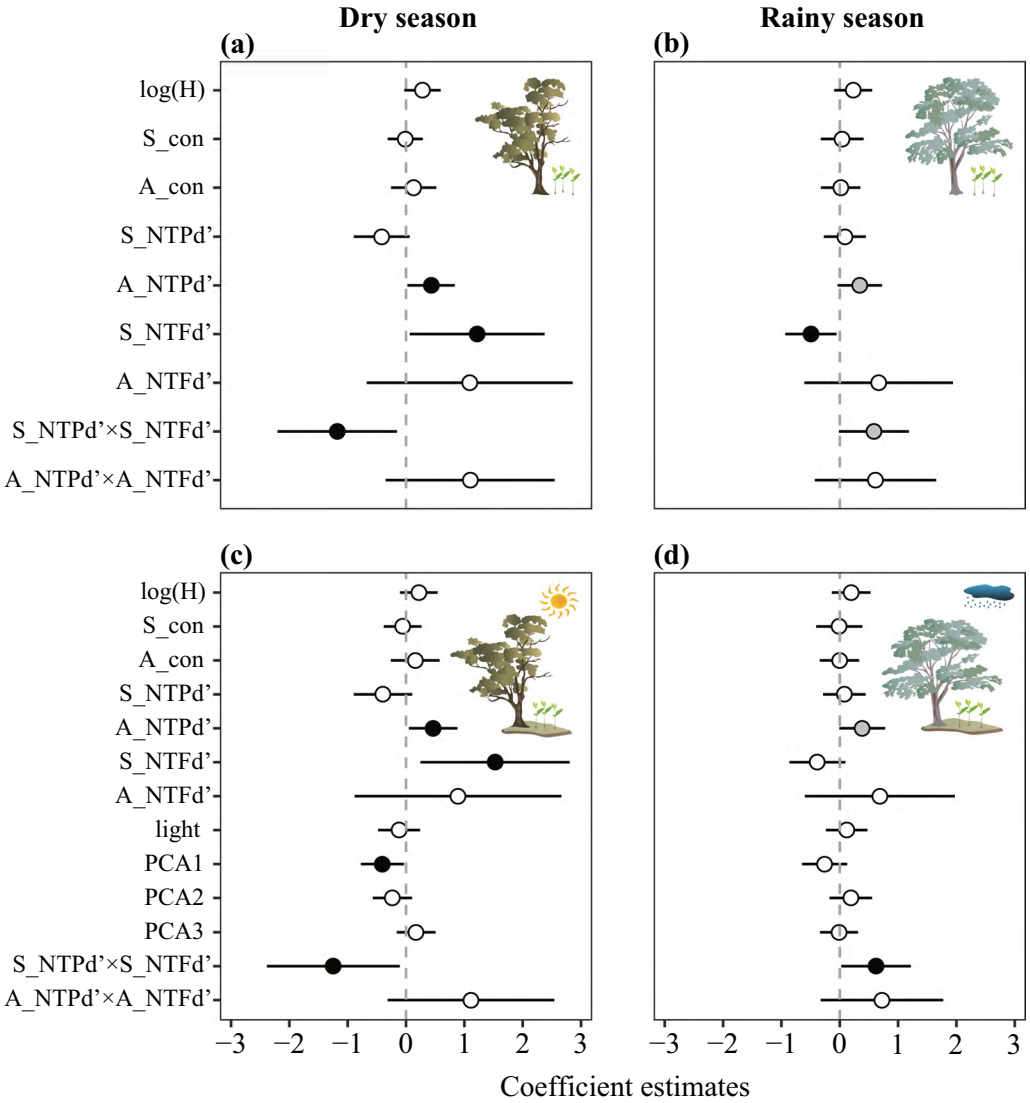


Fig. 2 Estimated effects (mean ± 2 SE) of neighborhood and habitat variables on seedling survival in the dry and rainy seasons. **a**, **b** Model I (without habitat variables); **c**, **d** Model II (with habitat variables). Bars indicate 95% confidence intervals. Black circles indicate significant effects ($P < 0.05$), gray circles indicate marginally significant effects ($0.05 < P < 0.1$), and white circles indicate non-significance effects ($P \geq 0.1$). See Table 1 for variable abbreviations

significant effect on seedling survival in either model. For habitat factors, only PCA1 negatively influenced seedling survival in the dry season (Table 1; Fig. 2c), an effect not observed in the rainy season. Besides, the strength of fixed effects was considerably higher in the dry season than in the rainy season.

Strikingly, the effects of PNDD, FNDD, and habitat heterogeneity on seedling survival were largely confined to the dry season, with PNDD effects becoming stronger after accounting for habitat variables. These findings suggest that habitat filtering can seasonally mask the detection of density-dependent mechanisms, highlighting the importance of integrating habitat heterogeneity and seasonal context into studies of forest community dynamics.

The masking effect of habitat filtering altered the interaction between phylogenetic and functional density dependence across seasons

In Model I, the interaction between phylogenetic and functional distance among seedling neighbors (S_NTPd'×S_NTFd') had a significant negative effect on seedling survival in the dry season but a significant positive effect in the rainy season (Table 1; Fig. 2a, b). When habitat variables were included in Model II, the

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interaction effects were strengthened in both seasons relative to Model I, with stronger effects observed in the dry season than in the rainy season (Table 1; Fig. 2c, d).

Specifically, in the dry season, seedlings that were phylogenetically close to their neighbors had the highest survival when surrounded by functionally dissimilar neighbors, whereas phylogenetically distant seedlings survived best when their neighbors were functionally similar (Fig. 3a). A similarly strong interaction was detected in the rainy season, but with opposite patterns: seedlings that were phylogenetically close to their neighbors survived best when their neighbors were functionally similar, whereas phylogenetically distant seedlings survived best when their neighbors were functionally dissimilar (Fig. 3c).

Consistent with these patterns, the slope of the relationship between S_NTPd' and seedling survival (logit scale) switched from positive to negative along the S_NTFd' gradient in the dry season (Fig. 3b), and from negative to positive in the rainy season (Fig. 3d). In general, the results from the GLMMs and partial derivative approaches were consistent across seasons, confirming the robustness of the S_NTPd'×S_NTFd' interaction in both dry and rainy seasons.

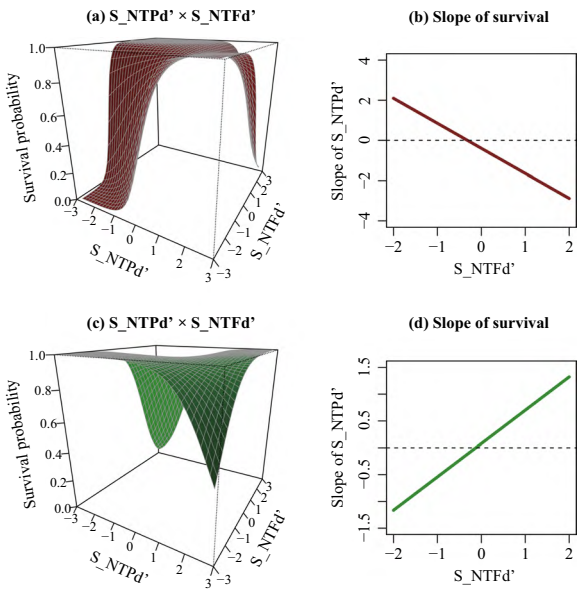


Fig. 3 Interaction effect between phylogenetic distance and functional distance among seedling neighbors (S_NTPd'×S_NTFd') on seedling survival in the dry (red) and rainy (green) seasons. Results are based on significant interactions from Model II. Panels **a**, **c** show response surfaces of seedling survival to S_NTPd' and S_NTFd'. In the dry season (**a**), survival is highest for seedlings that are phylogenetically close to their neighbors when surrounded by functionally dissimilar neighbors, and for phylogenetically distant seedlings when surrounded by functionally similar neighbors. In the rainy season (**c**), the opposite pattern is observed. Panels **b**, **d** illustrate how the slope of the relationship between S_NTPd' and logit-transformed survival changes along the S_NTFd' gradient using partial derivatives; the dotted line indicates a slope of zero. The slope crosses the zero line in both seasons, indicating a strong interaction

Seasonal variation in the masking effect at the species level

Because A_NTPd' and S_NTFd' showed significant or marginally significant effects on seedling survival (Table 1), we added species-specific random slopes for these predictors to Model I and Model II to examine species-level variation in the masking effect. The species-specific coefficients for A_NTPd' differed significantly between Model I and Model II in both the dry season ($t = -6.904$, $P < 0.001$; Fig. 4; Table S3) and the rainy season ($t = -2.900$, $P < 0.01$; Fig. 4; Table S3). For S_NTFd', a significant difference between models was detected in the dry season ($t = -19.808$, $P < 0.001$; Fig. 4; Table S3). However, there was no significant inter-model variation for S_NTFd' in the rainy season ($t = -1.882$, $P = 0.063$; Fig. 4; Table S3).

When habitat variables were included in Model II, the strength of PNDD increased for 92.31% of focal species in the dry season and for 96.15% in the rainy season, relative to Model I. Similarly, the effect of FNDD was stronger in Model II for 100% of species in the dry season and for 98.08% in the rainy season.

Discussion

Seasonal variation of density dependence and habitat filtering on seedling survival

The effect of CNDD was not detected in our study, potentially due to the relatively low density of conspecific neighbors compared to other forests such as Barro Colorado Island (Chen et al. 2018; Zhu et al. 2015). Additionally, adequate resource availability may contribute to the absence of CNDD, as CNDD is often weaker under high resource conditions where resource competition is alleviated (e.g., LaManna et al. 2017; Wright 2002). Future work should further explore how interactions between resource availability (e.g., soil fertility, moisture) and conspecific density influence seedling survival, as these interactions mediate the strength of density-dependent processes. In contrast, phylogenetic distance among neighbors had a significant positive effect on seedling survival, indicating a clear role of PNDD. These results are consistent with previous studies showing that focal seedling survival increases in the presence of distantly related neighbors (Metz et al. 2010; Paine et al. 2012; Webb et al. 2006; Zhu et al. 2015). Lower seedling survival with closer related neighbors may result from a greater likelihood of sharing natural enemies and pathogens (Chen et al. 2018; Liu et al. 2012; Weiblen et al.

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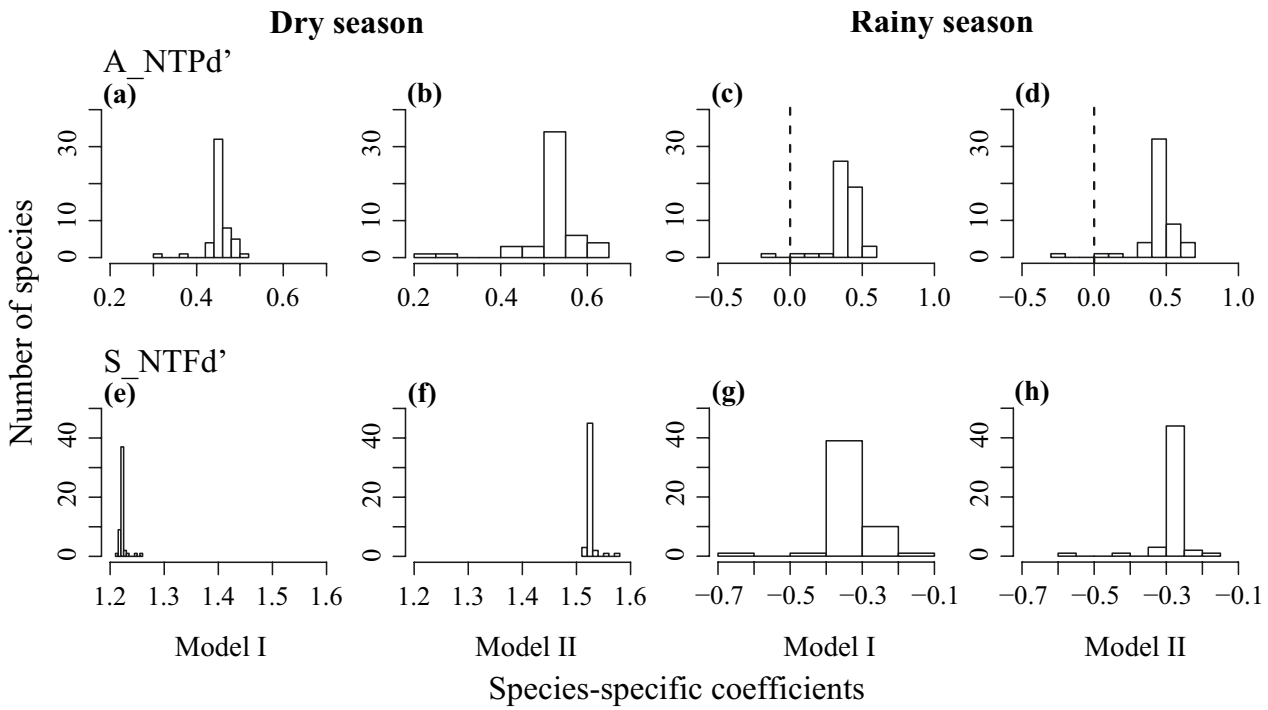


Fig. 4 Comparison of the frequency distribution of species-specific coefficients of A_NTPd' (**a**–**d**) and S_NTFd' (**e**–**h**) between Model I (without habitat variables) and Model II (with habitat variables) across dry and rainy seasons. Dashed vertical lines separate negative and positive coefficients. Positive coefficients indicate that species survival increases with higher factor values, whereas negative coefficients indicate the opposite

2006). However, some studies have reported phylogenetic positive density dependence (PPDD) in other forests, due to shared habitat preferences between closely related individuals (Huang et al. 2022; Lebrija-Trejos et al. 2014; Wang et al. 2024a; Wu et al. 2016). Furthermore, the stronger PNDD effects observed during the dry season likely reflects intensified resource competition under water-limited conditions. When water becomes scarce, closely related neighbors—which often share similar resource acquisition strategies—may compete more intensely, reducing seedling survival (Lin et al. 2012). In contrast, during the rainy season, reduced water stress may alleviate such competitive interactions, weakening the detectable signal of PNDD.

Limiting similarity theory posits that stable coexistence among competitors requires niche differentiation, as species with high ecological similarity are unlikely tocoexist (MacArthur and Levins 1967). Consistent with this theory, we observed FNDD in the dry season, supporting the view that niche differences are mediated by functional traits (Lasky et al. 2014, 2015). Accordingly, neighbors with lower trait similarity (reflecting greater niche differentiation) tend to exert weaker competitive effects (Pu et al. 2020). In contrast to the dry season, seedling survival was not influenced by functional similarity during the rainy season. This seasonal variation aligns with Laughlin et al. (2018), who reported that trait effects on survival depended on environmental conditions, with no trait exhibiting independent main effects on survival. For instance, leaf traits can vary substantially with soil water availability (Zhang et al. 2023). Similarly, evidence from tree species along the lowland Panama Isthmus indicates that changes in leaf traits such as SLA and LT are driven by variation in both soil phosphorus and moisture (Umaña et al. 2021). Leaf total K was also found to be lower in the dry season than in the rainy season (Zhang et al. 2023). Therefore, our findings further confirm that functional dissimilarity among neighbors is seasonally dependent. The absence of FNDD effects in the rainy season, despite increased pathogen activity expected under moist conditions (Lebrija-Trejos et al. 2023), suggests that functional similarity may influence seedling survival primarily through resource competition rather than shared natural enemies. When water is abundant, competition for other resources may be weaker, allowing functionally similar neighbors to coexist without measurable negative effects on survival (Lasky et al. 2014).

The negative effect of elevation on seedling survival was consistent with Yao et al. (2020) and Weng et al. (2022). Most seedling quadrats in our study were located on slopes, where elevation and convexity reflect changes in microhabitats and positional

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attributes. Given the relatively low rainfall in the dry season, soil moisture was generally reduced at higher elevations, likely reducing water availability for seedlings in these locations or at slope tops. The significant negative relationship between SOM and seedling survival in the dry season suggests that heavier litter input (e.g., from Fagaceae leaves) might harm seedling survival, concealing the potential benefits of higher soil fertility (Luo et al. 2024). The stronger habitat filtering effects in the dry season are consistent with the idea that abiotic stress amplifies the importance of environmental sorting (Kraft et al. 2015). Under drought conditions, seedlings must possess traits that enable survival in specific microhabitats (e.g., higher elevations with lower soil moisture), making habitat filters more selective. In the rainy season, relaxed water stress allows a broader range of species to survive across microhabitats, diminishing the detectable signal of habitat filtering. Taken together, these results indicate that habitat effects on seedling survival vary substantially across seasons.

The direction of the interaction between phylogenetic relatedness and functional similarity varied seasonally. On the one hand, the negative interaction effect observed during the dry season supports Darwin's naturalization hypothesis (Daehler 2001), which posits that more phylogenetically and functionally closely related neighbors decrease seedling survival. On the other hand, the interaction effect between phylogenetic and functional distance was associated with decreased seedling survival in the rainy season. In this context, closer phylogenetic relatedness to neighbors may be beneficial for seedling survival, supporting the pre-adaptation hypothesis (Ricciardi and Mottiar 2006). This suggests that closely related species share similar traits that facilitate adaptation to common environmental conditions (Ricciardi and Mottiar 2006). Consequently, our results demonstrate the importance of considering both phylogenetic and functional dimensions to fully explain variation in seedling survival (Cadotte et al. 2013). Together, these seasonal patterns suggest that the relative importance of density-dependent and habitat-based mechanisms is strongly modulated by water availability. During the dry season, resource limitation intensifies competition among functionally similar neighbors and strengthens habitat filtering, while the rainy season relaxes these constraints, dampening the signals of PNDD, FNDD, and habitat heterogeneity. This interpretation highlights the need to integrate seasonal climate variability into models of forest community dynamics, as the mechanisms driving seedling survival may shift dramatically across seasons.

Habitat filtering affects the detections of phylogenetic and functional density dependence and their interactions on seedling survival

Phylogenetic density dependence is a well-established mechanism driving seedling dynamics and coexistence (Metz et al. 2010; Webb et al. 2006; Wu et al. 2016; Zhu et al. 2015). However, the influence of habitat filtering on its detectability has rarely been considered in evergreen broad-leaved forests. This oversight is critical because previous studies have shown that failing to account for habitat filtering can lead to underestimation of CNDD (Cao et al. 2018; Wu et al. 2016). Our results confirm and extend this finding to phylogenetic and functional dimensions: when habitat variables were included in the models, the effects of PNDD became significantly stronger in both seasons, and the explained variance increased (Table 1). Moreover, habitat filtering similarly masked the detection of FNDD during the dry season. These results indicate that the positive effects of habitat filtering on seedling survival partially offset the negative effects of phylogenetic relatedness and functional similarity.

The interactive effects of phylogenetic relatedness and functional similarity on seedling survival were also amplified in models that incorporated habitat factors. This finding underscores that the importance of phylogenetic relatedness and functional similarity for species coexistence becomes more pronounced when habitat filtering shapes community assembly (Götzenberger et al. 2016). Furthermore, the inclusion of habitat variables consistently intensified PNDD and FNDD for most species in both seasons, indicating that the masking effect occurs at the species level as well. This provides mechanistic insight into why functional traits alone often fail to predict tree performance (Poorter et al. 2018; Wright et al. 2010), as important habitat factors (e.g., light and soil water gradients) may obscure trait–performance relationships (Li et al. 2022). The predictive ability of functional traits is substantially enhanced once these environmental contexts are controlled.

Overall, these results demonstrate that habitat filtering modulates the effect of phylogenetic and functional density dependence on seedling survival. As a key driver of local community structure, habitat filtering acts to mask the observed strength of these density-dependent processes. Furthermore, the interactions between phylogenetic and functional density dependence likely serve as a stronger driver of seedling survival at the community and species levels than either process in isolation. In summary, our findings indicate that while both habitat filtering and neighbor density dependence shape community composition, the former can mask the full influence of the latter.

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Conclusion

This study demonstrates that habitat filtering masks the detection of phylogenetic and functional density dependence on seedling survival, with this masking effect varying seasonally and operating consistently across both community and species levels. Crucially, accounting for habitat heterogeneity reveals significant interactions between phylogenetic and functional density dependence that are otherwise obscured, and these interactions shift direction between dry and rainy seasons. These findings reveal that habitat filtering fundamentally shapes the expression of biotic interactions across seasons, highlighting the necessity of integrating habitat heterogeneity and seasonal dynamics into studies of forest community assembly.

Abbreviations

PCA1	The first principal component
PCA2	The second principal component
PCA3	The third principal component
NDD	Negative density dependence
PNDD	Phylogenetic negative density dependence
FNDD	Functional negative density dependence
S_con	Conspecific neighbor densities of seedling
A_con	Conspecific neighbor densities of adult
S_NTPd'	Relative nearest taxon phylogenetic distance between heterospecific seedling neighbors and focal seedlings
A_NTPd'	Relative nearest taxon phylogenetic distance between heterospecific adult neighbors and focal seedlings
S_NTFd'	Relative nearest taxon functional distance between heterospecific seedling neighbors and focal seedlings
A_NTFd'	Relative nearest taxon functional distance between heterospecific adult neighbors and focal seedlings

Supplementary Information

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Additional file1 (DOCX 38 kb)

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Author contributions

Liping Wang conceived and designed the study, collected the field samples, analyzed the data, prepared figures and tables, wrote the original manuscript, reviewed the manuscript, and approved the final draft. Junjie Wu supervised the study and reviewed the manuscript. Yong Chai contributed the adult tree and topographic data. Fengxian Chen assisted in collecting the field samples.

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Availability of data and materials

Data is available at the Dryad Digital Repository https://doi.org/10.5061/dryad.nk98sf80r (Wang et al. 2024b).

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All authors approved the manuscript for publication in the Ecological Processes.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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